ERGA Assembly Report
v24.04.03_beta

Tags: ERGA-BGE

TxID	70637
ToLID	**heEubCras1**
Species	Eubothrium crassum
Class	Cestoda
Order	Bothriocephalidea

Genome Traits	Expected	Observed
Haploid size (bp)	3,663,021,826	1,219,776,824
Haploid Number	7 (source: ancestor)	8
Ploidy	2 (source: ancestor)	2
Sample Sex	U	U

EBP metrics summary and curation notes

Obtained EBP quality metric for pri: 6.8.Q64

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Haploid size (bp) has >20% difference with Expected
. Observed Haploid Number is different from Expected

. Kmer completeness value is less than 90 for pri
. BUSCO single copy value is less than 90% for pri
. Assembly length loss > 3% for pri

Curator notes

. Interventions/Gb: 165
. Contamination notes: "Contamination report for assembly labelled hap1; Total length of scaffolds removed: 2,396,831,736 (63.8 %); Scaffolds removed: 7036 (83.0 %); Largest scaffold removed: (112,143,520); FCS-GX contaminant species (number of scaffolds; total length of scaffolds): Salmo salar, fishes (7018; 2,395,005,402); Esox lucius, fishes (12; 258,103); Lawsonia intracellularis, d-proteobacteria (1; 1,437,186); Thalassophryne amazonica, fishes (1; 42,837); Eimeria tenella, alveolates (1; 33,525); Carcharodon carcharias, fishes (1; 12,000); Mitochondrion (2; 42,616); Barcodes (4; 67); Abnormal contamination report: in assembly hap1 TOTAL_LENGTH_REMOVED is 2396831736 which is above the alarm threshold 10042616.0. Abnormal contamination report: in assembly hap1 PERCENTAGE_LENGTH_REMOVED is 63.82010695465526 which is above the alarm threshold 3.001134730334687. Abnormal contamination report: in assembly hap1 LARGEST_SCAFFOLD_REMOVED is 112143520 which is above the alarm threshold 2000000.0. Abnormal contamination report: in assembly hap1 PERCENTAGE_SCAFFOLDS_REMOVED is 82.97169811320755 which is above the alarm threshold

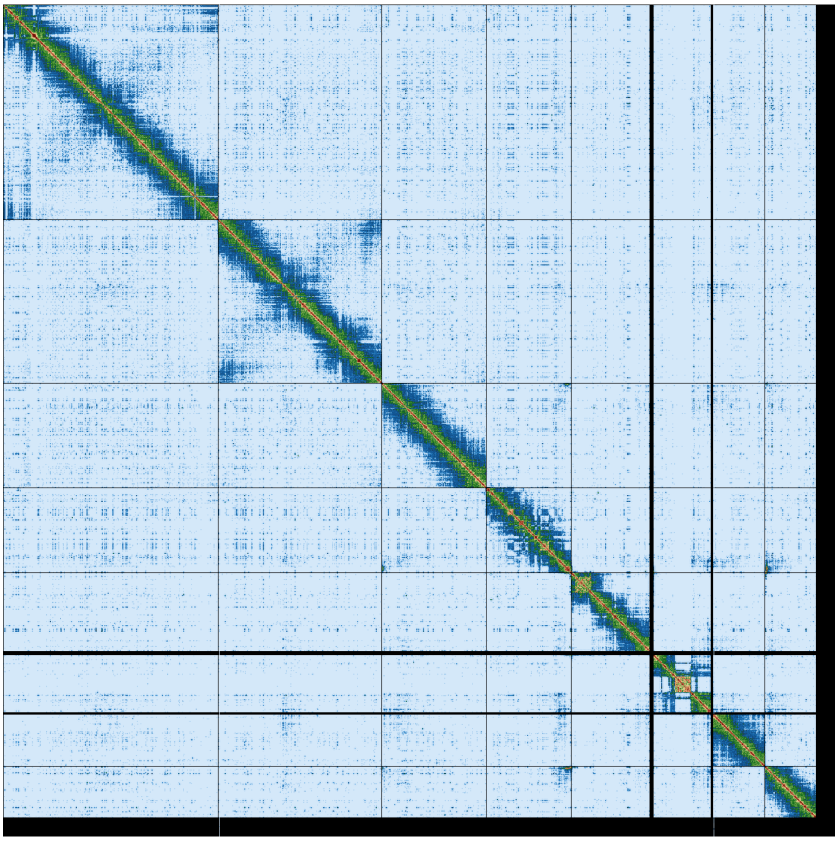
10.023584905660377. FCS-GX assigns scaffold HAP1_SCAFFOLD_73 ambiguous verdict REVIEW which requires manual review; FCS-GX assigns scaffold HAP1_SCAFFOLD_78 ambiguous verdict REVIEW which requires manual review; FCS-GX assigns scaffold HAP1_SCAFFOLD_100 ambiguous verdict REVIEW which requires manual review; FCS-GX assigns scaffold HAP1_SCAFFOLD_125 ambiguous verdict REVIEW which requires manual review; FCS-GX assigns scaffold HAP1_SCAFFOLD_170 ambiguous verdict REVIEW which requires manual review; FCS-GX assigns scaffold HAP1_SCAFFOLD_220 ambiguous verdict REVIEW which requires manual review; FCS-GX assigns scaffold HAP1_SCAFFOLD_454 ambiguous verdict REVIEW which requires manual review; FCS-GX assigns scaffold HAP1_SCAFFOLD_1129 ambiguous verdict REVIEW which requires manual review"
. Other observations: "A significant amount of the sequence data is from the fish this organism was parasitising, hence the low kmer completeness and QV when using the raw HiFi reads as the kmer database. Hifiasm run in Hi-C phasing mode; The exact order and orientation of the contigs on chromosome 5 (12.54-27.08 Mbp) and chromosome 6 (36.48-52.82Mbp) is unknown. The region on chromosome 5 (82.4-90.9 Mbp) exists in the same topology on hap2, so it was left as is. 8 chomosomes matches https://doi.org/10.1023/A:1011904911543 rather than the 7 from GoaT."

Quality metrics table

Metrics	Pre-curation pri	Curated pri
Total bp	3,755,605,953	1,219,776,824
GC %	42.21	40.02
Gaps/Gbp	1,033.39	966.57
Total gap bp	388,100	143,800
Scaffolds	8,480	604
Scaffold N50	58,067,801	153,026,447
Scaffold L50	18	3
Scaffold L90	649	7
Contigs	12,361	1,783
Contig N50	1,340,000	1,944,000
Contig L50	742	195
Contig L90	3,860	702
QV	61.9	64.9
Kmer compl.	98.81	23.10
BUSCO sing.	39.1%	68.2%
BUSCO dupl.	60.5%	3.4%
BUSCO frag.	0.2%	2.8%
BUSCO miss.	0.2%	25.6%

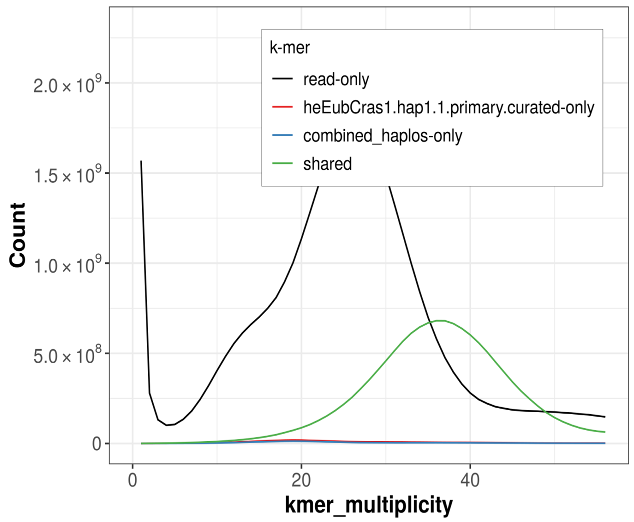
BUSCO 6.0.0 Lineage: metazoa_odb10 (genomes:65, BUSCOs:954)

HiC contact map of curated assembly

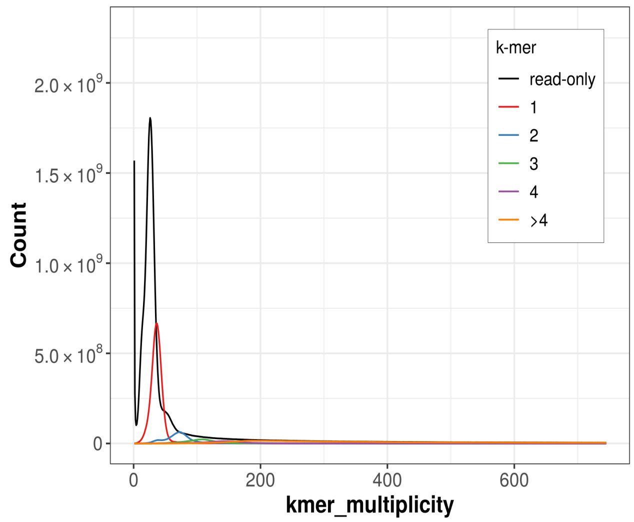


pri [LINK]

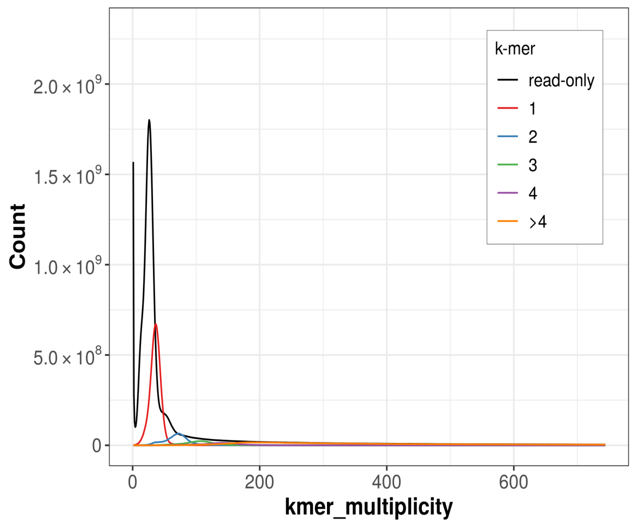
K-mer spectra of curated assembly



Distribution of k-mer counts coloured by their presence in reads/assemblies

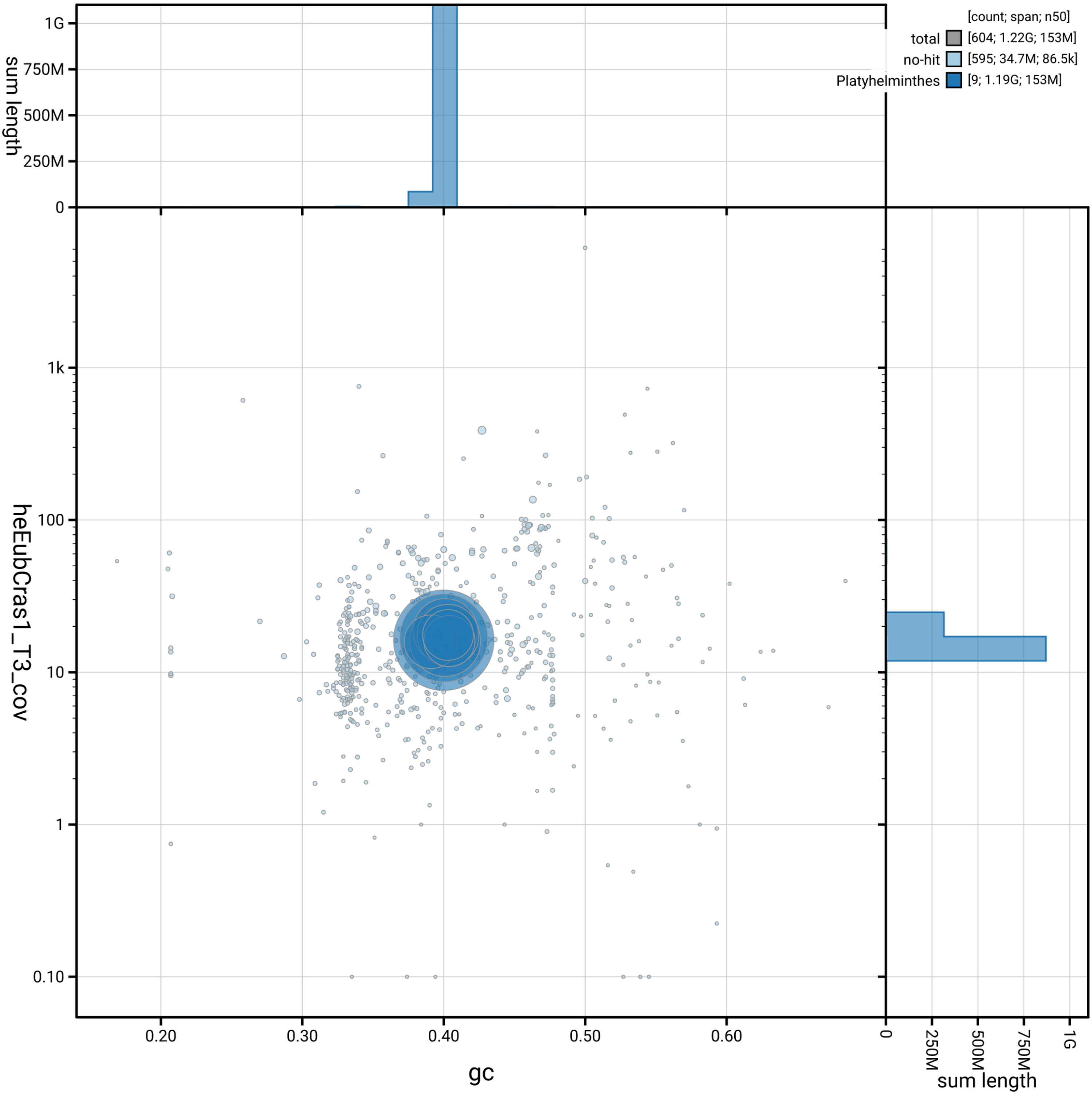


Distribution of k-mer counts per copy numbers found in asm



Distribution of k-mer counts per copy numbers found in asm

Post-curation contamination screening



pri. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	PacBio HiFi	Arima v2
Coverage	27x	31x

Assembly pipeline

```
- hifiasm
    |_ ver: 0.25.0-r726
    |_ key param: --h1/--h2
- purge_dups
    |_ ver: 1.2.5
    |_ key param: NA
- yahs
    |_ ver: 1.2.2
    |_ key param: NA
```

Curation pipeline

```
- hifiasm
    |_ ver: 0.25.0-r726
    |_ key param: --h1/--h2
- purge_dups
    |_ ver: 1.2.5
    |_ key param: NA
- yahs
    |_ ver: 1.2.2
    |_ key param: NA
- TreeVal
    |_ ver: 1.2.1
    |_ key param: NA
```

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Date and time: 2026-08-07 12:40:34 CEST